



SYSTEM DESIGN DOCUMENT
WAH4HOSPITAL (WAH4H)
Asia Pacific College
May-6 2026





INTRODUCTION

This System Design Document (SDD) describes the architecture, components, data models, security controls, and technical specifications for the WAH4H Hospital Information System developed by Team Pseudoers (BSCS-SS231) for the Wireless Access for Health (WAH) organization.

WAH4H is a full-stack, containerized web application that digitizes the complete clinical and administrative workflow of small municipal hospitals — from patient registration, admission, and monitoring through discharge, billing, pharmacy dispensing, laboratory management, appointment scheduling, and PhilHealth eClaims processing. The system is built in compliance with Philippine health informatics standards including PH Core FHIR R4, PhilHealth eClaims protocols, ICD-10-PH, LOINC, SNOMED CT, PSGC, MIMS PH, and RVS.

The current deliverable is a demonstration version — a fully functional proof-of-concept intended for presentation to local government officials (governors and mayors) to secure institutional buy-in and funding for full production deployment. The system is deployed on WAH's on-premise Proxmox Virtual Environment via Docker, secured through Cloudflare Tunnel.

This document is intended as a reference for the development team, WAH's Technical Supervisor (Sir Myk), and academic evaluators at Asia Pacific College.

PURPOSE

The purpose of this SDD is to:

- Define the technical architecture and component design of WAH4H in a form accessible to WAH's technical team for post-handover maintenance and future development.
- Communicate design decisions and their rationale — including technology stack choices, FHIR R4 compliance approach, deployment pipeline design, and security architecture — to all project stakeholders.
- Serve as the authoritative technical reference for UAT validation, compliance verification, and the Technical Manual produced as part of the project turnover package.
- Satisfy the System Design Document requirement under WBS 1.2.3 and the academic deliverable requirements of APC BSCS-SS231 PROJMAN.

This SDD covers all 11 modules across PBL1 and PBL2 phases: Accounts, Patients, Admission, Monitoring, Discharge, Billing, Pharmacy, Laboratory, Admin, Appointments, and eClaims.



SYSTEM OVERVIEW

WAH4H is a modular, role-based Hospital Information System built on a modern three-tier web architecture. The system transitions WAH's hospital operations from manual, paper-based workflows to a fully digital, standards-compliant platform.

6.1.3.1 System Goals

- Replace manual clinical workflows with a unified digital platform covering the complete patient journey.
- Achieve full PH Core FHIR R4 compliance at the data layer across all clinical resources.
- Enable automated PhilHealth eClaims processing, reducing billing errors and reimbursement delays.
- Provide role-specific interfaces for 7 hospital user roles with enforced RBAC at both UI and API layers.
- Deploy via Docker on WAH's Proxmox server, eliminating recurring cloud hosting costs (₱97,632/year equivalent savings vs. Linode).
- Maintain data sovereignty under RA 10173 through fully on-premise deployment.

6.1.3.2 Module Summary

#	Module	Primary Function
1	Accounts	User account management, JWT authentication, RBAC for 7 roles, role-specific dashboards
2	Patients	Patient registration, PH Core FHIR R4 Patient resource, PSGC address coding, PhilHealth number
3	Admission	Patient admission workflow, FHIR Encounter resource, ICD-10-PH admitting diagnosis, ward/room assignment
4	Monitoring	Vital signs (FHIR Observation + LOINC), nursing notes, physician orders, intake/output tracking
5	Discharge	Discharge processing, FHIR Encounter disposition, ICD-10-PH discharge diagnosis, clearance generation
6	Billing	Itemized billing, FHIR Invoice/ChargeItem, PhilHealth/HMO deductions, official receipt generation
7	Pharmacy	Medication dispensing, FHIR MedicationRequest/Administration, MIMS PH drug coding, stock management
8	Laboratory	Lab test management, FHIR DiagnosticReport/Observation, LOINC-coded tests, result release workflow
9	Admin	Facility/staff management, FHIR Organization/Location/PractitionerRole, audit logs
10	Appointments	Outpatient scheduling, FHIR Appointment resource, calendar views, booking/rescheduling/cancellation



11	eClaims	PhilHealth eClaims, FHIR Claim/ClaimResponse, CF1/CF2/CF4 forms, RVS coding, batch submission
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6.1.3.3 Technology Stack

Layer	Technology
Backend Framework	Django 5.x (Python) + Django REST Framework
Frontend Framework	React 18 + Vite + TailwindCSS + Zustand (state) + React Query (data fetching)
Database	PostgreSQL 15+
Authentication	JSON Web Tokens (JWT) via djangorestframework-simplejwt; RBAC enforced via custom DRF permission classes
Containerization	Docker + Docker Compose (pinned base images for environment parity)
Hypervisor	Proxmox VE (on-premise; WAH's physical server)
Secure Tunnel	Cloudflare Tunnel (external HTTPS access without exposed server ports)
Security (Server)	Fail2Ban (brute-force protection on Proxmox host)
Operating System	Linux (Ubuntu 22.04 LTS on Proxmox VMs)
Version Control	Git / GitHub

DESIGN CONSTRAINTS

The following constraints are fixed boundaries that directly shaped design decisions throughout WAH4H development.

Constraint	Design Impact
WAH Proxmox Hardware Limits	System must be deployable on 16GB RAM, 320GB storage, 8 CPUs. All Docker resource allocations and container counts are sized within these limits. No vertical scaling available.
PH Core FHIR R4 Compliance	All clinical resources (Patient, Encounter, Observation, Condition, Procedure, MedicationRequest, DiagnosticReport, Claim, etc.) must conform to Philippine FHIR R4 profiles. Schema designed FHIR-first before feature development.
RA 10173 (Data Privacy Act)	Patient data must be encrypted at rest and in transit. All access must be logged. No data may leave the WAH infrastructure without authorization. Drives on-premise Proxmox deployment decision.
Demo Version Scope	Current deliverable is a demonstration version only. Architecture must support future production-grade scaling without requiring a full redesign.



No External SSL / Cloud Routing	External SSL certificates and cloud-based routing are out of scope. Cloudflare Tunnel provides secure HTTPS access using WAH's existing \$20/month subscription.
Fixed Team Capacity	4-member team. No additional resources. Architecture must be maintainable and deployable by a small team; Docker Compose simplifies the full-stack deployment to a single command.
PhilHealth / NHFR Prerequisites	PhilHealth accreditation and NHFR registration are WAH responsibilities. eClaims module is architected demo-ready; live eClaims batch submission is gated on WAH completing accreditation.

ROLES AND RESPONSIBILITIES

Name	Role	System Design Responsibility
Jhon Lloyd Nicolas	Lead Developer	System architecture, Django backend, FHIR R4 compliance, API design, module-level data design, security implementation (JWT/RBAC/audit logging)
Mariyah Vanna Monique C. Chavez	Quality Assurance / DevOps	Proxmox/Docker infrastructure design, deployment pipeline, Cloudflare Tunnel config, Fail2Ban config, environment parity validation, PHCore R4 alignment verification
Elijah Josh R. Quibin	Project Manager	Design traceability to WBS/Charter, stakeholder communication on design decisions, acceptance criteria alignment, document control
John Kenneth L. Jajurie	Scrum Master / Agile Developer	Frontend UI architecture (React component structure, role-based routing, Zustand state design), sprint-level feature delivery
Sir Myk (Monte)	Technical Supervisor, WAH	WAH-side infrastructure review, deployment environment decisions, WAH4P/WAH4PC interoperability specification, source code and Technical Manual acceptance
Sir Robert Michael Martinez	Server Manager, WAH	Proxmox server provisioning, VM network bridging, production infrastructure management

PROJECT REFERENCES

Ref ID	Document / Standard	Description
REF-01	Project Charter v1.0, May 2026	Authorizes WAH4H scope, deliverables, stakeholders, budget, and milestone schedule.
REF-02	PH Core FHIR R4 Implementation Guide	Philippine adaptation of HL7 FHIR R4 governing all clinical resource structures in WAH4H.



REF-03	PhilHealth eClaims Protocol (CF1/CF2/CF4)	Electronic claims submission format and form field requirements for PhilHealth reimbursement.
REF-04	ICD-10-PH (Philippine edition)	Diagnosis coding standard used in Admission, Monitoring, Discharge, and eClaims modules.
REF-05	LOINC (Logical Observation Identifiers)	Coding standard for laboratory tests and clinical observations in Laboratory and Monitoring modules.
REF-06	SNOMED CT	Clinical terminology for conditions, procedures, and findings in Monitoring and Discharge modules.
REF-07	MIMS PH (Drug Reference)	Philippine drug reference applied to medication coding in the Pharmacy module.
REF-08	RVS (Relative Value Scale)	Procedure cost coding system used in PhilHealth eClaims billing.
REF-09	PSGC (Philippine Standard Geographic Code)	Standardized geographic coding applied to all patient and facility address fields.
REF-10	RA 10173 — Data Privacy Act of 2012	Governs all patient data handling, encryption, access logging, and audit trails.
REF-11	RA 11223 — Universal Health Care Act	Governs PhilHealth interoperability and UHC alignment for eClaims and Billing modules.
REF-12	WAH4H WBS v1.1, May 2026	Full work breakdown structure governing all project deliverables and design artifacts.
REF-13	WAH4H Requirements Management Plan, May 2026	Requirements traceability matrix and product metrics governing design acceptance criteria.
REF-14	Django 5.x Documentation	Official framework documentation for backend implementation.
REF-15	Proxmox VE Documentation	Official documentation governing on-premise hypervisor configuration.

SYSTEM ARCHITECTURE

6.1.7.1 Deployment Architecture Overview

WAH4H uses a three-tier deployment pipeline: Local Developer Environment → Proxmox Sandbox VM → Proxmox Production VM. All tiers are Docker-based with pinned base images to guarantee environment parity. In this document, Production VM means the WAH-hosted demo-ready deployment environment, not the final clinical production go-live environment.

Tier	Environment	Purpose
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1	Local (4× Developer Machines)	Docker-based development and unit testing. Environment parity via pinned base images. No direct access to WAH infrastructure.
2	Proxmox Sandbox VM	Integration testing and full-system validation. Mirrors production configuration exactly. PM must approve promotion to Production.
3	Proxmox Production VM	Demo-ready deployment. Cloudflare Tunnel enabled for secure external access. Served to WAH staff and government demonstration audiences.

6.1.7.2 Application Architecture

WAH4H follows a decoupled frontend/backend architecture with a REST API communication layer. The Django backend exposes a fully stateless JWT-authenticated API consumed by the React frontend.

Component	Description
React Frontend (Port 3000)	Single-page application built with React 18 + Vite. TailwindCSS for styling. Zustand for global state management (auth context, user role, current patient). React Query for server state, caching, and API calls. Role-based routing enforces module access per RBAC definitions.
Django REST API (Port 8000)	Django 5.x with Django REST Framework. JWT authentication via SimpleJWT middleware. Custom RBAC permission classes enforced at the view layer. FHIR R4-aligned serializers per module. Audit logging middleware captures all write operations automatically.
PostgreSQL (Port 5432)	Primary data store. Schema designed to PH Core FHIR R4 profiles. Django ORM manages migrations. All FK constraints enforced at DB level for data integrity. Regular automated backups as described in the Technical Manual.
Nginx Reverse Proxy	Sits in front of both Django and the React static build. Routes /api/ requests to Django and serves the React SPA for all other paths. Handles static file serving and gzip compression.
Cloudflare Tunnel (cloudflared)	Provides secure HTTPS external access without opening server ports. Tunnels traffic through Cloudflare's network to the Nginx reverse proxy inside the Production VM. Replaces the need for an external SSL certificate.
Fail2Ban	Installed on the Proxmox host. Monitors authentication logs and bans IPs after 5 failed login attempts within 60 seconds. Protects both SSH and application-layer access.

6.1.7.3 Hardware Architecture



WAH4H is deployed on WAH's existing physical server managed by Sir Robert Michael Martinez (Server/Deployment Manager, WAH). No hardware procurement is required under this project.

Specification	Detail
Server Type	Bare-metal physical server managed by WAH
RAM	16 GB
Storage	320 GB
CPUs	8 Cores
Operating System	Linux (Ubuntu 22.04 LTS)
Hypervisor	Proxmox VE (latest stable)
Network	VM network bridging configured by Sir Robert Michael Martinez. Cloudflare Tunnel provides external access. Internal LAN for WAH staff access.

6.1.7.4 Docker Compose Stack Design

All WAH4H components are containerized via Docker Compose. The following services are defined in docker-compose.yml with pinned image versions:

Service	Base Image	Role
backend	python:3.11-slim	Django 5.x REST API + Gunicorn WSGI server
frontend	node:20-alpine (build)	React 18 + Vite build; static files served by Nginx
db	postgres:15-alpine	PostgreSQL 15 database
nginx	nginx:alpine	Reverse proxy; routes /api/ to backend, serves React SPA
cloudflared	cloudflare/cloudflared:latest	Cloudflare Tunnel daemon — secure external access

DATABASE DESIGN

6.1.8.1 Design Principles

- Schema is designed FHIR-first: all clinical entities are modeled against PH Core FHIR R4 resource profiles before any feature development begins.
- All FK constraints are enforced at the PostgreSQL level — not just at the application layer — to guarantee referential integrity.
- Django ORM manages migrations. All migration files are version-controlled and included in the Source Code Repository turnover package.
- Sensitive fields (patient identifiers, contact details) are encrypted at the application layer before storage (RA 10173 compliance).



6.1.8.2 Core Data Model — FHIR Resource Mapping

Module	Primary FHIR Resource(s)	Key Fields
Patients	Patient	id, identifier (PhilHealth), name, birthDate, gender, address (PSGC), telecom, maritalStatus, bloodType, emergencyContact
Admission	Encounter	id, status, class, type, subject (→Patient), participant (Practitioner), period, location (→Location), reasonCode (ICD-10-PH), account (→Account)
Monitoring	Observation	id, status, category, code (LOINC), subject (→Patient), encounter (→Encounter), effectiveDateTime, valueQuantity, component
Discharge	Encounter (updated)	hospitalization.dischargeDisposition, hospitalization.diagnoses (ICD-10-PH), period.end, status=finished
Billing	Invoice + ChargeItem	Invoice: id, subject, total, lineItem; ChargeItem: code, factorOverride (PhilHealth/HMO deduction), priceOverride
Pharmacy	MedicationRequest + MedicationAdministration	MedRequest: medication (MIMS PH code), subject, requester, dosage; MedAdmin: medication, subject, occurrenceDateTime, dosage
Laboratory	DiagnosticReport + Observation	DiagReport: code (LOINC), subject, result (→Observations), status; Observation: code, value, specimen
Admin	Organization + Location + PractitionerRole	Org: name, identifier (NHFR); Location: name, type, managingOrganization; PractRole: practitioner, organization, code, specialty
Appointments	Appointment	id, status, serviceType, specialty, start, end, participant (→Patient, →PractitionerRole)
eClaims	Claim + ClaimResponse	Claim: patient, provider (NHFR), diagnosis (ICD-10-PH), procedure (RVS), insurance, item; ClaimResponse: outcome, payment, adjudication

6.1.8.3 Reference Tables

The following reference tables are populated before any module enters UAT. Incomplete reference tables are a blocking issue per QM-02 in the Quality Management Plan.

Table	Standard	Applied In
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icd10_ph_codes	ICD-10-PH	Admission (admitting diagnosis), Monitoring, Discharge, eClaims
loinc_codes	LOINC	Laboratory (test codes), Monitoring (vital sign Observation codes)
snomed_ct_codes	SNOMED CT	Monitoring (clinical findings), Discharge (condition terminology)
mims_ph_drugs	MIMS PH	Pharmacy (medication coding, dispensing records)
psgc_codes	PSGC 2023	Patients (address region/province/city-municipality/barangay), Admin (facility address)
rvs_codes	RVS	eClaims (procedure cost coding), Billing (procedure billing)

HARDWARE AND SOFTWARE DETAILED DESIGN

6.1.9.1 Hardware

WAH4H leverages WAH's existing physical server. No hardware procurement is within Team Pseudoers' scope. Hardware capital expenditure is a WAH responsibility managed by Sir Robert Michael Martinez and documented in the Cost Management Plan (Section 3.6.4.3).

Recommended WAH hardware investment (post-demo phase):

- UPS (Uninterruptible Power Supply) — P1 priority; protects against brownouts common in Philippine municipal settings.
- Redundant ISP (secondary broadband line + dual-WAN router) — P1 priority; ensures uptime during primary ISP failures.
- Managed Gigabit Network Switch — P1 priority; enables VLAN segmentation for security isolation.
- Secondary/Backup Physical Server (Proxmox Backup Server) — P2; high-availability failover.

6.1.9.2 Backend — Django REST Framework

The Django backend is organized into Django apps corresponding to each module. The following design patterns are applied consistently:

Pattern	Implementation
URL Routing	DRF Router auto-generates RESTful endpoints. Each module app has its own urls.py registered in the project-level urls.py under /api/v1/<module>/.
Serializers	FHIR-aligned serializers per resource type. Custom validators enforce code constraints (LOINC, ICD-10-PH, MIMS PH) at serializer level before DB write.



ViewSets	ModelViewSet for standard CRUD. Custom actions (@action) for domain-specific operations (e.g., result_release on DiagnosticReport, discharge_patient on Encounter).
Permissions	Custom DRF permission classes per role per module. Enforces RBAC at the API layer independently of the frontend role-based routing.
Audit Middleware	Django middleware intercepts all POST, PUT, PATCH, DELETE requests and writes an audit log entry: user ID, timestamp (UTC), method, resource type, record ID, HTTP response status.
CORS	django-cors-headers configured to allow the React frontend origin only. No wildcard CORS.
Environment Config	All secrets (SECRET_KEY, DB credentials, JWT secret) stored in .env files. django-environ parses environment variables. No secrets committed to the repository.

6.1.9.3 Frontend — React 18

The React frontend is organized by module, with a shared component library for common UI elements (forms, data tables, status badges). Key design decisions:

- Role-based routing: React Router v6 protects module routes. A route guard component checks the authenticated user's role against a role-permission map. Unauthorized route access redirects to the user's home dashboard.
- Zustand stores: AuthStore (JWT token, user profile, role), PatientContextStore (active patient ID across module views).
- React Query: Handles all API data fetching, caching, and mutation. Automatic cache invalidation on write operations ensures consistent data display.
- TailwindCSS + custom tokens: UI color system derived from WAH4H design: primary blue #0099FF, accent purple #46017F, orange #FF9100.
- Vite: Used as the build tool for fast hot module replacement in development and optimized production bundle generation.

6.1.9.4 API Design Conventions

Convention	Detail
Base URL	/api/v1/
Authentication Header	Authorization: Bearer <access_token>
Token Expiry	Access token: 60 minutes. Refresh token: 7 days. Refresh via POST /api/v1/auth/token/refresh/.
Response Format	JSON. Success: { data: {...}, message: '...' }. Error: { error: '...', detail: '...' }. Pagination: { count, next, previous, results }.



HTTP Status Codes	200 OK, 201 Created, 204 No Content, 400 Bad Request (validation), 401 Unauthorized, 403 Forbidden (RBAC), 404 Not Found, 500 Internal Server Error.
FHIR Resource Output	Module-specific endpoints return FHIR-compliant JSON representations. A /fhir/ endpoint namespace exposes FHIR-native resource endpoints for interoperability.

COMPLIANCE ARCHITECTURE

6.1.10.1 PH Core FHIR R4 Implementation Approach

PH Core FHIR R4 compliance is established at the database schema layer before any feature development begins. The compliance architecture follows a four-track approach:

- Gap Analysis — Review of existing module schemas against PH Core FHIR R4 profiles to identify unmapped fields and missing resource linkages.
- PHCore R4 Schema Alignment — Database migration to align all clinical tables with PH Core resource field structures.
- Field Validation — Serializer-level validation enforcing FHIR data types, cardinalities, and code system constraints.
- Reference Table Population — All required coding reference tables (ICD-10-PH, LOINC, SNOMED CT, MIMS PH, PSGC, RVS) populated to 100% completeness before corresponding modules enter UAT.

6.1.10.2 Cross-System Interoperability (WAH4P / WAH4PC)

WAH4H must synchronize patient and clinical data with the WAH4P (patient-facing) and WAH4PC (community/portal) platforms. The integration architecture is as follows:

- The system design includes a dedicated interoperability service layer intended to handle outbound sync events once the WAH technical specification is finalized.
- Sync events are triggered by specific write operations (e.g., patient registration, admission, discharge) and dispatch FHIR-compliant payloads to WAH4P/WAH4PC API endpoints.
- Cross-platform FHIR resource references (e.g., Patient.id) are maintained as stable identifiers across all three platforms.

SYSTEM SECURITY AND INTEGRITY CONTROLS

6.1.11.1 Authentication — JWT

All WAH4H API endpoints are protected by JWT authentication implemented via `djangorestframework-simplejwt`. Authentication flow:



- User submits credentials (username + password) to POST /api/v1/auth/token/.
- Django validates credentials, generates access token (60-min expiry) and refresh token (7-day expiry), returns both.
- Frontend stores tokens in memory (AuthStore). Access token sent as Authorization: Bearer header on every API request.
- On access token expiry, frontend automatically calls POST /api/v1/auth/token/refresh/ with the refresh token to obtain a new access token.
- Expired or absent tokens return HTTP 401. Role-unauthorized requests return HTTP 403.

6.1.11.2 Role-Based Access Control (RBAC)

Seven hospital roles are defined. Each role has a defined permission set enforced at both the Django API layer (DRF permission classes) and the React frontend (route guards + conditional rendering).

#	Role	Primary Access Scope
1	Hospital Admin	Full system access. Audit log view. User management. System configuration.
2	Doctor	Patient records, Admission, Monitoring (orders + view), Discharge, Appointments (own schedule), Laboratory (result view), eClaims (own patients)
3	Nurse	Monitoring (vitals, nursing notes, I&O), Admission (view), Discharge (view), Pharmacy (dispense confirmation)
4	Pharmacist	Pharmacy (full: prescriptions, dispensing, stock management, low-stock alerts)
5	Lab Tech	Laboratory (full: test requests, specimen tracking, result entry, result release)
6	Billing Staff	Billing (full: invoicing, PhilHealth/HMO deductions, OR generation), eClaims (submission workflow)
7	Receptionist/Admin	Patients (registration, search), Appointments (booking, rescheduling, cancellation), Admission (intake processing)

6.1.11.3 Audit Logging

All clinical and financial write operations generate a tamper-evident audit log entry. Audit logs are stored in a dedicated audit_logs table with the following fields:

- user_id — authenticated user who performed the action
- timestamp — UTC timestamp of the operation
- http_method — POST / PUT / PATCH / DELETE
- resource_type — affected FHIR resource type (e.g., Encounter, Observation, Invoice)



- `record_id` — primary key of the affected record
- `http_status` — response status code
- `ip_address` — client IP address

Audit logs are readable only by the Hospital Admin role. The audit log view endpoint returns HTTP 403 for all other roles.

6.1.11.4 Data Encryption

Patient-identifiable fields are encrypted at rest using Django field-level encryption (via `django-encrypted-model-fields` or equivalent). All external traffic is encrypted in transit via Cloudflare Tunnel (TLS 1.2+). No patient data is transmitted over unencrypted channels.

6.1.11.5 Infrastructure Security

Control	Implementation
Cloudflare Tunnel	Sole external entry point. No application ports (8000, 5432, 80) are exposed to the public internet. All external traffic routes through Cloudflare's network.
Fail2Ban	Bans an IP after 5 failed login attempts within 60 seconds. Configured on the Proxmox host monitoring <code>/var/log/auth.log</code> and application access logs.
Environment Separation	No code change is deployed directly to Production without first passing Sandbox validation. PM must provide written approval for each Sandbox-to-Production promotion.
Secret Management	All secrets (<code>SECRET_KEY</code> , DB credentials, JWT secret, Cloudflare tunnel token) stored in <code>.env</code> files. Never committed to the Git repository. Separate <code>.env</code> files per deployment tier.
Database Access	PostgreSQL port (5432) accessible only from the Django backend container via Docker network. Not exposed to the host OS or external network.

6.1.11.6 Backup and Recovery

Automated daily PostgreSQL dumps are stored off-server. Full restore procedures are documented in the Technical Manual (WBS 1.9.3) and tested in the Production environment before formal handoff. Proxmox VM snapshots provide additional point-in-time recovery capability.

Sponsor Acceptance

Approved by the Project Sponsor:

Date:



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Executive Director, WAH

